



Editorial

Novel anaerobic taxa and taxonomic revisions published in 2024, and requirements for the validation process to establish taxon names with standing in the literature

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Microbial taxonomy, which deals with the identification, classification, and nomenclature of microorganisms, underpins the foundation of all microbiology. The principles that support taxonomy remain remarkably stable; however, theoretical and methodological advances used to assign taxa to classification frameworks continue to evolve and influence taxonomy [1–4].

The nomenclature aspect involves assigning scientifically correct and internationally recognized names to different microbial taxa delineated by the classification methods. The International Code for Nomenclature of Prokaryotes (the ICNP or ‘the Code’) embodies the principles, rules, and recommendations for the naming of archaeal and bacterial taxa, including rules for the valid publication of names and establishing priority in the literature [5]. However, this system only caters for organisms recovered in culture; with the recognition that most organisms cannot be isolated in culture, Murray and Schleifer proposed a new category of taxonomic name, which they called *Candidatus*, based on sequence-based recognition of new taxa [6]. More recently, with the explosion of classification of new uncultured taxa primarily through analysis of metagenome-assembled genome sequences (MAGs) or single-cell amplified genomes (SAGs) [7,8] resulted in a call for the Code to give standing to uncultured taxa [9–11]. Proposals were made to augment the International Code of Nomenclature of Prokaryotes (ICNP) [9,10] to allow for DNA sequences (mainly, but not only, genome sequences) to serve as nomenclatural types of species or subspecies, in addition to cultivable, viable type strains.

However, after much debate and a vote by their members, the International Committee on Systematics of Prokaryotes (ICSP) rejected that proposal [12], thus leading to the formation of an alternative code of nomenclature for uncultured microbes called the Code of Nomenclature of Prokaryotes Described from Sequence Data (known as SeqCode) [13–16]. Organisms described within the SeqCode can be registered at the SeqCode Registry (<https://registry.seqco.de/>). Still, it is

pertinent to recognize that the status ‘validly published under the SeqCode’ must not be confused with the bona fide validation of names processed through the IJSEM. It is important to note that naming taxa under the SeqCode does not and should not prevent efforts to isolate representatives of taxa and deposit strains in culture collections. In fact, information contained within whole genome sequences of uncultivated taxa should stimulate efforts to isolate representatives of these taxa for further investigation. The availability of organisms in culture collections is still of great value for the modern description of prokaryotes and their metabolic capabilities. Even when its genome is known, the current state-of-the-art tools poorly predict phenotypic properties; laboratory experiments on living organisms are still necessary, especially in microbiome studies dealing with complex ecosystems and microbe-microbe and microbe-host interactions [17–19].

To keep our readership abreast of novel anaerobes published in the scientific literature, the Anaerobe journal is introducing a new feature: each year, a list of novel, strictly anaerobic bacteria published in the scientific literature from the previous year will be listed as an Editorial section (see Table 1). While compiling this for the organisms published in 2024, it became apparent that many had not been validly published in IJSEM. What is this process, and what does it mean practically? When a novel species description is published in a scientific journal other than IJSEM, this is known as the “Effective publication” along with the date associated with the publication, but the name itself has no standing in the literature until it appears in the Validation Lists published in the IJSEM. The procedure itself is straightforward: all that is needed is to submit to the IJSEM editorial office (ijsem@microbiologysociety.org) an electronic copy of the final published version in which the names were published, in addition (for species and subspecies) documentation that the type strains are available from two culture collections in accordance with the requirements of the Code [5,20,21]. The editorial office staff and the List Editors will further process the requests. If the requirements

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Table 1

New anaerobic bacterial species reported in 2024.

Scientific name	Family	Source	Reference
<i>Adlercreutzia faecimuris</i> sp. nov.	<i>Eggerthellaceae</i>	Mouse faeces	M.K. Suh, J.-S. Jin, H.E. Do, J.-S. Kim, M.K. Eom, H.S. Kim, J.-S. Lee, <i>Adlercreutzia faecimuris</i> sp. nov., producing propionate and acetate isolated from mouse faeces, <i>Antonie van Leeuwenhoek</i> 117 (2024) 80. https://doi.org/10.1007/s10482-024-01980-y .
^a <i>Aequorivita ciconiae</i> sp. nov.	<i>Flavobacteriaceae</i>	Faecal sample of oriental stork	J.-E. Han, S. Kang, S.-Y. Lee, J.-W. Bae, Characterisation of <i>Aequorivita ciconiae</i> sp. nov., isolated from oriental stork, <i>Ciconia boyciana</i> , <i>Antonie van Leeuwenhoek</i> 118 (2025) 47. https://doi.org/10.1007/s10482-024-02056-7 .
<i>Anaerobaca lacustris</i> gen. nov., sp. nov.	<i>Anaerobacaceae</i>	Coastal lake	M.A. Khomyakova, A.Y. Merkel, A.I. Slobodkin, <i>Anaerobaca lacustris</i> gen. nov., sp. nov., an obligately anaerobic planctomycete of the widespread SG8-4 group, isolated from a coastal lake, and proposal of <i>Anaerobacaceae</i> fam. nov, <i>Syst. Appl. Microbiol.</i> 47 (2024) 126522. https://doi.org/10.1016/j.syapm.2024.126522 .
<i>Anaerococcus kampingiae</i> sp. nov. <i>Anaerococcus groningenensis</i> sp. nov. <i>Anaerococcus martiniensis</i> sp. nov. <i>Anaerococcus cruorum</i> sp. nov.	<i>Peptoniphilaceae</i>	Human clinical specimens	K.E. Boiten, J. Meijer, E.M. van Wezel, A.C.M. Veloo, Description of <i>Anaerococcus kampingiae</i> sp. nov., <i>Anaerococcus groningenensis</i> sp. nov., <i>Anaerococcus martiniensis</i> sp. nov., and <i>Anaerococcus cruorum</i> sp. nov., isolated from human clinical specimens, <i>Anaerobe</i> 91 (2025) 102935. https://doi.org/10.1016/j.anaerobe.2024.102935 .
<i>Anaerolentibacter hominis</i> gen. nov. sp. nov.	<i>Lachnospiraceae</i>	Human faeces	Y. Yan, Q. Zhao, C. Xu, R. Wei, H. Jiang, S. Liu, <i>Anaerolentibacter hominis</i> gen. nov. sp. nov., <i>Diplocloster hominis</i> sp. nov. and <i>Pilosibacter fragilis</i> gen. nov. sp. nov., isolated from human faeces, <i>Int. J. Syst. Evol. Microbiol.</i> 74 (2024). https://doi.org/10.1099/ijsem.0.006359 .
<i>Atrimonas thermophila</i> gen. nov., sp. nov.	<i>Atrimonadaceae</i>	Deep subsurface gas field	H. Kawamoto, M. Watanabe, H. Mochimaru, N. Nakahara, X.-Y. Meng, S. Sakamoto, K. Morinaga, T. Katayama, H. Yoshioka, N. Nomura, H. Tamaki, <i>Atrimonas thermophila</i> gen. nov., sp. nov., a novel anaerobic thermophilic bacterium of the phylum <i>Atribacterota</i> isolated from deep subsurface gas field and proposal of <i>Atrimonadaceae</i> fam. nov. within the class <i>Atribacteria</i> in the phylum <i>Atribacterota</i> , <i>Syst. Appl. Microbiol.</i> 47 (2024) 126515. https://doi.org/10.1016/j.syapm.2024.126515 .
<i>Azotosporobacter soli</i> gen. nov., sp. nov.	<i>Dendrosporobacteraceae</i>	Paddy soil	C.-J. Xie, L. Yao, R. Tang, S. Han, S. Yang, H. Alwathnani, C. Rensing, G.-H. Liu, S.-G. Zhou, <i>Azotosporobacter soli</i> gen. nov., sp. nov., a novel nitrogen-fixing bacterium isolated from paddy soil, <i>Antonie van Leeuwenhoek</i> 117 (2024) 79. https://doi.org/10.1007/s10482-024-01978-6 .
<i>Bacteroides vicugnae</i> sp. nov.	<i>Bacteroidaceae</i>	Alpaca faeces	S. Miller, M. Hendry, J. King, K. Sankaranarayanan, P.A. Lawson, <i>Bacteroides vicugnae</i> sp. nov. isolated from the fecal material of an alpaca, <i>Anaerobe</i> 88 (2024) 102862. https://doi.org/10.1016/j.anaerobe.2024.102862 .
<i>Bengtsoniella intestinalis</i> gen. nov., sp. nov.	<i>Oscillospiraceae</i>	Hindgut of the marine herbivorous fish	B. Pardesi, A.M. Robertson, E.M. Wollmuth, E.R. Angert, D.I. Rosendale, L. W. White, K.D. Clements, <i>Bengtsoniella intestinalis</i> gen. nov., sp. nov., a member of the family <i>Oscillospiraceae</i> , isolated from the hindgut of the marine herbivorous fish <i>Kyphosus sydneyanus</i> , <i>Int. J. Syst. Evol. Microbiol.</i> 74 (2024). https://doi.org/10.1099/ijsem.0.006615 .
<i>Clostridium aquiflavi</i> sp. nov.	<i>Clostridiaceae</i>	Water	S. Sun, H. Feng, R. Liu, J. Cheng, S. Mu, Y. Liu, Y. Gao, S. Yao, L. Zhai, <i>Clostridium aquiflavi</i> sp. nov., isolated from yellow water of Nongxiangxing baijiu in the Yibin region of China, <i>Int. J. Syst. Evol. Microbiol.</i> 74 (2024). https://doi.org/10.1099/ijsem.0.006420 .
<i>Clostridium lamae</i> sp. nov.	<i>Clostridiaceae</i>	Alpaca faeces	X. Li, M. Li, W. Shi, X. Li, Z. Xiang, L. Su, <i>Clostridium lamae</i> sp. nov., a novel bacterium isolated from the fresh faeces of alpaca, <i>Antonie van Leeuwenhoek</i> 117 (2024) 36. https://doi.org/10.1007/s10482-024-01931-7 .
<i>Clostridium omnivorum</i> sp. nov.	<i>Clostridiaceae</i>	Anoxic soil	A.Ueki, A. Tonouchi, N. Kaku, S. Honma, K. Ueki, <i>Clostridium omnivorum</i> sp. nov., isolated from anoxic soil under the treatment of reductive soil disinfection, <i>Int. J. Syst. Evol. Microbiol.</i> 74 (2024). https://doi.org/10.1099/ijsem.0.006412 .
<i>Clostridium tanneri</i> sp. nov.	<i>Clostridiaceae</i>	Alpaca faeces	Miller, M. Hendry, J. King, K. Sankaranarayanan, P.A. Lawson, <i>Clostridium tanneri</i> sp. nov., isolated from the faecal material of an alpaca, <i>Int. J. Syst. Evol. Microbiol.</i> 74 (2024). https://doi.org/10.1099/ijsem.0.006372 .
^a <i>Denitrificimonas halotolerans</i> sp. nov.	<i>Pseudomonadaceae</i>	UASB sludge treating landfill leachate,	S. Wang, C. Yuan, C. Xu, D. Li, H. Zhang, J. Wang, X. Wang, Y. Li, D. Jiao, S. Yuan, H. Chen, D. Qiu, <i>Denitrificimonas halotolerans</i> sp. nov., a novel species isolated from UASB sludge treating landfill leachate, <i>Antonie van Leeuwenhoek</i> 117 (2024) 91. https://doi.org/10.1007/s10482-024-01987-5 .
<i>Desulfofundulus salinus</i> sp. nov.	<i>Peptococcaceae</i>	Reservoir fluids from high-temperature hydrocarbon deposits	T.N. Nazina, T.P. Tourova, D.S. Grouzdev, S.K. Bidzhieva, A.B. Poltarau, A Novel View on the Taxonomy of Sulfate-Reducing Bacterium ‘ <i>Desulfotomaculum salinum</i> ’ and a Description of a New Species <i>Desulfofundulus salinus</i> sp. nov., <i>Microorganisms</i> 12 (2024) 1115. https://doi.org/10.3390/microorganisms12061115 .
<i>Diplocloster hominis</i> sp. nov.	<i>Lachnospiraceae</i>	Human faeces	Y. Yan, Q. Zhao, C. Xu, R. Wei, H. Jiang, S. Liu, <i>Anaerolentibacter hominis</i> gen. nov. sp. nov., <i>Diplocloster hominis</i> sp. nov. and <i>Pilosibacter fragilis</i> gen.

(continued on next page)

Table 1 (continued)

Scientific name	Family	Source	Reference
<i>Enterovibrio qingdaonensis</i> sp. nov.	Vibrionaceae	Marine surface water	nov. sp. nov., isolated from human faeces, Int. J. Syst. Evol. Microbiol. 74 (2024). https://doi.org/10.1099/ijsem.0.006359 .
<i>Enterovibrio gelatinilyticus</i> sp. nov.			K. Huang, Y. Liu, Y. Wang, P. Zhang, H. Cheng, J. Wang, W. Tong, W. Xiaolei, X.-H. Zhang, <i>Enterovibrio qingdaonensis</i> sp. nov. and <i>Enterovibrio gelatinilyticus</i> sp. nov., two marine bacteria isolated from surface seawater of Qingdao offshore, Antonie van Leeuwenhoek 118 (2025) 17. https://doi.org/10.1007/s10482-024-02030-3 .
<i>Eubacterium album</i> sp. nov.	Eubacteriaceae	Human faeces	F.-L. Liu, R. Abdugheni, C.-G. Ran, N. Zhou, S.-J. Liu, <i>Eubacterium album</i> sp. nov., a butyrate-producing bacterium isolated from faeces of a healthy human, Int. J. Syst. Evol. Microbiol. 74 (2024). https://doi.org/10.1099/ijsem.0.006380 .
<i>Finegoldia dalianensis</i> sp. nov.	Peptoniphilaceae	Pus from human skin abscess	Y. Li, Y. Wang, D. Xiao, J. Wang, D. Jin, The identification of <i>Finegoldia dalianensis</i> sp. nov., isolated from the pus of a patient with skin abscess and genomic analysis of the strains belonging to <i>Finegoldia</i> genus, Anaerobe 90 (2024) 102913. https://doi.org/10.1016/j.anaerobe.2024.102913 .
<i>Faecalibacterium taiwanense</i> sp. nov.	Ruminococcaceae	Human faeces	J.-S. Liou, W.-L. Zhang, L.-W. Hsu, C.-C. Chen, Y.-T. Wang, K. Mori, K. Hidaka, M. Hamada, L. Huang, K. Watanabe, C.-H. Huang, <i>Faecalibacterium taiwanense</i> sp. nov., isolated from human faeces, Int. J. Syst. Evol. Microbiol. 74 (2024). https://doi.org/10.1099/ijsem.0.006413 .
<i>Faecalibacterium wellingii</i> sp. nov.	Ruminococcaceae	Human faeces	N. Plomp, H.J.M. Harmsen, Description of <i>Faecalibacterium wellingii</i> sp. nov. and two <i>Faecalibacterium taiwanense</i> strains, aiding to the reclassification of <i>Faecalibacterium</i> species, Anaerobe 89 (2024) 102881. https://doi.org/10.1016/j.anaerobe.2024.102881 .
<i>Ferviditalea candida</i> gen. nov., sp. nov.	Paenibacillaceae	Soil, geothermal area	Y. Chen, A.-P. Lv, M.-M. Li, Y.-T. OuYang, Z.-H. Lian, L.-B. Chen, Z.-T. Liu, L. Liu, J.-Y. Jiao, W.-J. Li, <i>Ferviditalea candida</i> gen. nov., sp. nov., a novel member of the family <i>Paenibacillaceae</i> isolated from a geothermal area, Anaerobe 88 (2024) 102866. https://doi.org/10.1016/j.anaerobe.2024.102866 .
<i>Fidelibacter multiformis</i> gen. nov., sp. nov.	Fidelibacteraceae	Deep subsurface aquifer	T. Katayama, M.K. Nobu, Y. Kamagata, H. Tamaki, <i>Fidelibacter multiformis</i> gen. nov., sp. nov., isolated from a deep subsurface aquifer and proposal of <i>Fidelibacterota</i> phyl. nov., formerly called Marine Group A, SAR406 or Candidatus Marinimicrobia, Int. J. Syst. Evol. Microbiol. 74 (2024) 006558. https://doi.org/10.1099/ijsem.0.006558 .
<i>Fuscovulum ytuae</i> sp. nov.	Rhodobacteraceae	Intertidal zone sediment	Zhang, X. Xu, D. Zhou, X.-P. Li, X. Jiao, J. Zhang, Description of <i>Fuscovulum ytuae</i> sp. nov., a facultative autotroph isolated from the intertidalite of Yangma island, China, Antonie van Leeuwenhoek 117 (2024) 52. https://doi.org/10.1007/s10482-024-01947-z .
<i>Geothrix campi</i> sp. nov.	Holophagaceae	Paddy soils	S. Han, S. Yang, R. Tang, C.-J. Xie, X. Liu, G.-H. Liu, S.-G. Zhou, Two novel Fe(III)-reducing bacteria, <i>Geothrix campi</i> sp. nov. and <i>Geothrix mesophila</i> sp. nov., isolated from paddy soils, Antonie van Leeuwenhoek 117 (2024) 68. https://doi.org/10.1007/s10482-024-01967-9 .
<i>Halosquirtibacter laminarini</i> gen. nov., sp. nov.	Prolixibacteraceae	Sea squirt	T.T.H. Nguyen, T.Q. Vuong, H.L. Han, S.-G. Kim, <i>Halosquirtibacter laminarini</i> gen. nov., sp. nov. and <i>Halosquirtibacter xylanolyticus</i> sp. nov., marine anaerobic laminarin and xylan degraders in the phylum Bacteroidota, Sci. Rep. 14 (2024) 24329. https://doi.org/10.1038/s41598-024-74787-6 .
<i>Halosquirtibacter xylanolyticus</i> sp. nov.			
<i>Helicovermis profundus</i> gen. nov., sp. nov.	Acidaminobacteraceae	Deep-sea hydrothermal vent chimney,	U. Miyazaki, D. Mizutani, Y. Hashimoto, A. Tame, S. Sawayama, J. Miyazaki, K. Takai, S. Nakagawa, <i>Helicovermis profundus</i> gen. nov., sp. nov., a novel mesophilic, asporogenous bacterium within the <i>Clostridia</i> isolated from a deep-sea hydrothermal vent chimney, Antonie van Leeuwenhoek 117 (2024) 24. https://doi.org/10.1007/s10482-023-01919-9 .
<i>Kribbibacterium absianum</i> gen. nov., sp. nov.	Kribbibacteriaceae	Pig faecal dump	L. Bai, J. Paek, Y. Shin, H. Kim, S.H. Kim, J.H. Shin, J.-K. Kook, Y.-H. Chang, Description of an anaerobic actinobacterium, <i>Kribbibacterium absianum</i> gen. nov., sp. nov., a new member of the novel family <i>Kribbibacteriaceae</i> fam. nov., and reclassification of the genera <i>Granulimonas</i> and <i>Leptogranulimonas</i> , Int. J. Syst. Evol. Microbiol. 74 (2024). https://doi.org/10.1099/ijsem.0.006382 .
<i>Minisyncoccus archaeiphilus</i> gen. nov., sp. nov.	Minisyncoccaceae	Candidatus Patescibacteria- enriched culture in an anaerobic laboratory-scale bioreactor	M. Nakajima, R. Nakai, Y. Hirakata, K. Kubota, H. Satoh, M.K. Nobu, T. Narihiro, K. Kuroda, <i>Minisyncoccus archaeiphilus</i> gen. nov., sp. nov., a mesophilic, obligate parasitic bacterium and proposal of <i>Minisyncoccaceae</i> fam. nov., <i>Minisyncoccales</i> ord. nov., <i>Minisyncoccia</i> class. nov. and <i>Minisyncocota</i> phyl. nov. formerly referred to as <i>Candidatus Patescibacteria</i> or candidate phyla radiation, Int. J. Syst. Evol. Microbiol. 75 (2025). https://doi.org/10.1099/ijsem.0.006668 .
<i>Mobiluncus massiliensis</i> sp. nov.	Actinomycetaceae	Vaginal Microbiome	L.A. Chacra, M. Bonnet, M. Heredia, G. Haddad, N. Armstrong, S. Alibar, F. Bretelle, F. Fenollar, <i>Peptoniphilus genitalis</i> sp. nov. and <i>Mobiluncus massiliensis</i> sp. nov.: Novel Bacteria Isolated from the Vaginal Microbiome, Curr. Microbiol. 81 (2024) 97. https://doi.org/10.1007/s00284-023-03584-7 .
<i>Olsenella absiana</i> sp. nov.	Coriobacteriaceae	Pig faeces	J. Paek, L. Bai, Y. Shin, H. Kim, S.H. Kim, J.H. Shin, J.-K. Kook, Y.-H. Chang, Description of <i>Olsenella absiana</i> sp. nov., an anaerobic bacterium isolated from pig faeces, Int. J. Syst. Evol. Microbiol. 74 (2024). https://doi.org/10.1099/ijsem.0.006579 .

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Table 1 (continued)

Scientific name	Family	Source	Reference
<i>Paralibaculum antarcticum</i> gen. nov., sp. nov.	Marinifilaceae	Antarctica sea ice	Y. Zhuang, Y. Zhang, W. Dai, Y. Liang, X. Yang, Y. Wang, X. Shi, X.-H. Zhang, <i>Paralibaculum antarcticum</i> gen. nov., sp. nov., an anaerobic marine bacterium of the family Marinifilaceae isolated from Antarctica sea ice, <i>Antonie van Leeuwenhoek</i> 118 (2025) 8. https://doi.org/10.1007/s10482-024-02022-3 .
<i>Peloplasma aerotolerans</i> gen. nov., sp. nov.	Achleplasmataceae	Terrestrial mud volcano	M.A. Khomyakova, A.Y. Merkel, A.A. Novikov, A.I. Slobodkin, <i>Peloplasma aerotolerans</i> gen. nov., sp. nov., a Novel Anaerobic Free-Living Mollicute Isolated from a Terrestrial Mud Volcano, <i>Life</i> 14 (2024) 563. https://doi.org/10.3390/life14050563 .
<i>Peptostreptococcus equinus</i> sp. nov.	Peptoniphilaceae	Horse faeces	H.S. Jung, J.U. Yun, M.-J. Jung, H.S. Song, Y.B. Kim, Y. Kim, M.J. Lee, J. Lee, C.O. Jeon, S.W. Roh, T.W. Whon, J.-G. Kim, S.H. Lee, <i>Peptostreptococcus equinus</i> sp. nov., isolated from horse faeces, <i>Int. J. Syst. Evol. Microbiol.</i> 74 (2024). https://doi.org/10.1099/ijsem.0.006295 .
<i>Peptoniphilus genitalis</i> sp. nov.	Peptoniphilaceae	Vaginal microbiome	L.A. Chacra, M. Bonnet, M. Heredia, G. Haddad, N. Armstrong, S. Alibar, F. Bretelle, F. Fenollar, <i>Peptoniphilus genitalis</i> sp. nov. and <i>Mobiluncus massiliensis</i> sp. nov.: Novel Bacteria Isolated from the Vaginal Microbiome, <i>Curr. Microbiol.</i> 81 (2024) 97. https://doi.org/10.1007/s00284-023-03584-7 .
^a <i>Phocaeicola acetigenes</i> sp. nov.	Bacteroidaceae	Human faeces	H.E. Do, Y.B. Ha, J.-S. Kim, M.K. Suh, H.S. Kim, M.K. Eom, J.H. Lee, S.-H. Park, S.W. Kang, D.H. Lee, H. Yoon, J.H. Lee, J.-S. Lee, <i>Phocaeicola acetigenes</i> sp. nov., producing acetic acid and iso-butyric acid, isolated faeces from a healthy human, <i>Antonie van Leeuwenhoek</i> 117 (2024) 30. https://doi.org/10.1007/s10482-024-01930-8 .
<i>Pilosibacter fragilis</i> gen. nov. sp. nov.	Clostridiaceae	Human faeces	Y. Yan, Q. Zhao, C. Xu, R. Wei, H. Jiang, S. Liu, <i>Anaerolentibacter hominis</i> gen. nov. sp. nov., <i>Diplocloster hominis</i> sp. nov. and <i>Pilosibacter fragilis</i> gen. nov. sp. nov., isolated from human faeces, <i>Int. J. Syst. Evol. Microbiol.</i> 74 (2024). https://doi.org/10.1099/ijsem.0.006359 .
<i>Pontiella agarivorans</i> sp. nov.	Pontellaceae	Microbial mats on the surface of marine sediments	N. Liu, V. Kivenson, X. Peng, Z. Cui, T.S. Lankiewicz, K.M. Gosselin, C.J. English, E.M. Blair, M.A. O'Malley, D.L. Valentine, <i>Pontiella agarivorans</i> sp. nov., a novel marine anaerobic bacterium capable of degrading macroalgal polysaccharides and fixing nitrogen, <i>Appl. Environ. Microbiol.</i> 90 (2024) e00914-23. https://doi.org/10.1128/aem.00914-23 .
<i>Promethearchaeum syntrophicum</i> gen. nov., sp. nov.	Promethearchaeaceae	Deep-sea methane-seep sediment	H. Imachi, M.K. Nobu, S. Kato, Y. Takaki, M. Miyazaki, M. Miyata, M. Ogawara, Y. Saito, S. Sakai, Y.O. Tahara, Y. Takano, E. Tasumi, K. Uematsu, T. Yoshimura, T. Itoh, M. Ohkuma, K. Takai, <i>Promethearchaeum syntrophicum</i> gen. nov., sp. nov., an anaerobic, obligately syntrophic archaeon, the first isolate of the lineage 'Asgard', and proposal of the new archaeal phylum <i>Promethearchaeota</i> phyl. nov. and kingdom <i>Promethearchaeati</i> regn. nov., <i>Int. J. Syst. Evol. Microbiol.</i> 74 (2024). https://doi.org/10.1099/ijsem.0.006435 .
^a <i>Proteus appendicitidis</i> sp. nov.	Enterobacteriaceae	Appendiceal pus of an appendicitis patient	J. He, Z. Zhang, H. Qu, G. Chen, J. Zhou, W. Zhou, Y. Peng, X. Cai, <i>Proteus appendicitidis</i> sp. nov., isolated from the appendiceal pus of an appendicitis patient in Yongzhou, China, <i>Arch. Microbiol.</i> 206 (2024) 171. https://doi.org/10.1007/s00203-024-03887-8 .
^a <i>Pseudodesulfovibrio methanolicus</i> sp. nov.	Desulfovibrionaceae	Petroleum reservoir	S.K. Bidzhieva, T.P. Tourova, V.V. Kadnikov, S.R. Samigullina, D.S. Sokolova, A.B. Poltarau, A.N. Avtukh, V.M. Tereshina, A.V. Beletsky, A. V. Mardanov, T.N. Nazina, Phenotypic and Genomic Characterization of a Sulfate-Reducing Bacterium <i>Pseudodesulfovibrio methanolicus</i> sp. nov. Isolated from a Petroleum Reservoir in Russia, <i>Biology</i> 13 (2024) 800. https://doi.org/10.3390/biology13100800
^b <i>Pseudodesulfovibrio pelocollis</i> sp. nov.	Desulfovibrionaceae	Terrestrial mud volcano	G. Slobodkina, A. Merkel, A. Novikov, A. Slobodkin, <i>Pseudodesulfovibrio pelocollis</i> sp. nov. a Sulfate-Reducing Bacterium Isolated from a Terrestrial Mud Volcano, <i>Curr. Microbiol.</i> 81 (2024) 120. https://doi.org/10.1007/s00284-024-03644-6 .
<i>Selenomonas ruminis</i> sp. nov.	Selenomonadaceae	Rumen fluid of lactating goats	S. Poothong, S. Tanasupawat, S. Chanpongsang, E. Kingkaew, C. Nuengjamnong, Anaerobic flora, <i>Selenomonas ruminis</i> sp. nov., and the bacteriocinogenic <i>Ligilactobacillus salivarius</i> strain MP3 from crossbred-lactating goats, <i>Sci. Rep.</i> 14 (2024) 4838. https://doi.org/10.1038/s41598-024-54686-6 .
^a <i>Sinimarinibacterium thermocellulolyticum</i> sp. nov.	Nevskiaceae	Hot spring water	Y. Jiang, Z. Li, Comprehensive genomic and phenotypic characterization of thermophilic bacterium <i>Sinimarinibacterium thermocellulolyticum</i> sp. nov. HSW-8T, a cellulase-producing bacterium isolated from hot spring water in South Korea, <i>Antonie van Leeuwenhoek</i> 118 (2025) 45. https://doi.org/10.1007/s10482-024-02055-8 .
<i>Terrisporobacter vanillatitrophus</i> sp. nov.	Peptostreptococcaceae	Compost	T. Böer, M.A. Schüller, A. Lüschen, L. Eysell, J. Dröge, M. Heinemann, L. Engelhardt, M. Basen, R. Daniel, A. Poehlein, Isolation and characterization of novel acetogenic strains of the genera <i>Terrisporobacter</i> and <i>Acetoanaerobium</i> , <i>Front. Microbiol.</i> 15 (2024) 1426882. https://doi.org/10.3389/fmicb.2024.1426882 .
<i>Thermodesulfovibrio autotrophicus</i> sp. nov. <i>Thermodesulfovibrio obliviosus</i> sp. nov.	Thermodesulfovibrionaceae	Hot spring	A.I. Maltseva, A.G. Elcheninov, A.A. Klyukina, N.V. Pimenov, A.A. Novikov, A.V. Lebedinsky, E.N. Frolov, <i>Thermodesulfovibrio autotrophicus</i> sp. nov., the first autotrophic representative of the widespread sulfate-reducing genus <i>Thermodesulfovibrio</i> , and <i>Thermodesulfovibrio obliviosus</i> sp. nov. that has lost this ability, <i>Syst. Appl. Microbiol.</i> 47 (2024) 126561. https://doi.org/10.1016/j.syapm.2024.126561 .

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Table 1 (continued)

Scientific name	Family	Source	Reference
<i>Thiobacillus sedimenti</i> sp. nov.	Thiobacillaceae	Freshwater sediment	C. Dai, G. Zhang, W. Lin, J. Luo, <i>Thiobacillus sedimenti</i> sp. nov., a chemolithoautotrophic sulphur-oxidizing bacterium isolated from freshwater sediment, Antonie van Leeuwenhoek 118 (2025) 9. https://doi.org/10.1007/s10482-024-02026-z .
<i>Thiopseudomonas acetoxidans</i> sp. nov.	Pseudomonadaceae	Anaerobic fermentation liquid of food waste	M. An, R. Liang, Y. Lu, X. Li, G. Zhao, <i>Thiopseudomonas acetoxidans</i> sp. nov., an aerobic acetic and butyric acids oxidizer isolated from anaerobic fermentation liquid of food waste, Antonie van Leeuwenhoek 117 (2024) 35. https://doi.org/10.1007/s10482-024-01932-6 .
<i>Turicibacter faecis</i> sp. nov.	Turicibacteraceae	Mouse faeces	Y. Imamura, D. Motooka, Y. Nakajima, S. Ito, M. Kitakaze, T. Iida, S. Nakamura, <i>Turicibacter faecis</i> sp. nov., isolated from faeces of heart failure mouse model, Int. J. Syst. Evol. Microbiol. 74 (2024) 006379. https://doi.org/10.1099/ijsem.0.006379 .
^a <i>Veillonella faecalis</i> sp. nov.	Veillonellaceae	Human faeces	H. Yang, H. Jo, S.H. Kim, C.-S. Yun, S.-H. Park, D.-S. Park, <i>Veillonella faecalis</i> sp. nov., a propionic acid-producing bacterium isolated from the faeces of an infant, Antonie van Leeuwenhoek 117 (2024) 50. https://doi.org/10.1007/s10482-024-01951-3 .
^a <i>Veillonella orientalis</i> sp. nov.	Veillonellaceae	Human saliva	I. Mashima-Usami, C.F. Theodora, K. Murata, B. Thaweeboon, S. Thaweeboon, F. Nakazawa, an anaerobic Gram-stain-negative coccus isolated from saliva of a Thai child, Anaerobe 91 (2025) 102921. https://doi.org/10.1016/j.anaerobe.2024.102921 .
<i>Xiashengella succiniciproducens</i> gen. nov.	Marinilabiliaceae	Anaerobic digestion tank	H.-J. Huang, X. Zhang, X.-W. Sun, B. Chen, X.-T. Li, N. Zhou, R. Abdugheni, Q.-Y. Cheng, T.-J. Zhang, Y. Liu, Y. Jiang, Y. Deng, S.-J. Liu, C.-Y. Jiang, <i>Xiashengella succiniciproducens</i> gen. nov., sp. nov., a succinate-producing bacterium isolated from an anaerobic digestion tank in the family Marinilabiliaceae of the order Bacteroidales, Arch. Microbiol. 206 (2024) 141. https://doi.org/10.1007/s00203-024-03909-5 .

^a Not validly published under the ICNP at the time of writing.

^b Aerotolerant anaerobe.

have been met, the names will appear in the Validations lists [22]. The exact process is used for the pro-validation of *Candidatus* names and published in *Candidatus* Lists, first published in IJSEM in 2020, and are now regularly updated [23]. Without being validly published, the names have no standing in the nomenclature/literature. In a historical review, Oren et al. [22] reported that between 2014 and 2017 nearly half the names published outside the IJSEM were never submitted for validation! Once validly published the names are incorporated into the List of Prokaryotic names with Standing in Nomenclature (LPSN; <https://lpsn.dsmz.de/>), the service founded by Jean P. Euzéby in 1997 and later maintained by Aiden C. Pate. As of February 2020, the LPSN database is administered through DSM and as of December 2023, LPSN is recognized as a Global Core Biodata Resource (<https://globalbiodata.org/>). LPSN is a valuable resource, and we encourage our readers to utilize this service to verify the status of species that may be included in their investigations. In addition to the IJSEM and LPSN, another excellent resource is the Genome Taxonomy Database (<https://gtdb.ecogenomic.org/>), which has become an important reference source for classification and genome-based taxonomy of *Bacteria* and *Archaea* that are based on objective evolutionary relationships [3,24–26].

In addition to our readers interested in anaerobes, many have a clinical interest. While taxonomy may not be a primary issue, it is important to be aware of novel organisms causing infections or the reclassification of well-established taxa that can cause confusion amongst clinicians, as exemplified by *Clostridium difficile* to *Clostridioides difficile* [27,28] and *Propionibacterium acnes* subsp. *acnes* to *Cutibacterium acnes* subsp. *acnes* [29]. Of course, the name change for any clinically meaningful organism has the potential to cause some confusion, but these should be outweighed by the benefits of a strong evolutionary-based taxonomy [24,30,31]. Furthermore, it is beneficial when authors of potentially controversial publications take care to explain the justification for the taxonomic revisions proposed [27,28,32].

Finally, it is pertinent to note that the purpose and goal of the LPSN database are based on nomenclature, rather than providing clinical information. However, a subcommittee of the ICSP, entitled the Ad Hoc Committee for Mitigating Changes in Prokaryotic Nomenclature (CoMiCProN), was formed in early 2024 by a group of medical and veterinary

researchers and taxonomic nomenclature experts. As a result, from 2025, LPSN has implemented several features useful to the clinical community. In addition to the Nomenclatural status (validly published or not), the Taxonomic status (including the correct name and explicit recommendation for medical use) and the Risk Group are now included. Furthermore, the following statement is now present for organisms of clinical relevance: “This name is on the List of Recommended Names for bacteria of medical importance (LoRN) as the name to be applied to this species. The name is on LoRN because of the recorded risk group of the species itself”. Some names are now placed in taxonomic suspension, while considered acceptable from a taxonomic viewpoint, their use is postponed for practical reasons (<https://lpsn.dsmz.de/text/taxonomic-suspension>). These will become the correct and recommended LoRN name when approved by the CoMiCProN and the ICSP. For information on novel organisms recovered from human clinical specimens and taxonomic revisions, the reader is directed to several excellent publications of Munson and Carroll [33–36] and Janda [37–39], respectively (note, these are not necessarily strict anaerobes).

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